

Experiment 3.1

Force between Two Electric Charges with Varying Distance

Code File Name: Exp3_1_ChargeForce_VaryingDistance.m

Code

```
clc;
clear;
close all;

q1 = input('Enter the charge q1 = ');
q2 = input('Enter the charge q2 = ');

epsilon = 8.854e-12;
k = 1/(4*pi*epsilon);

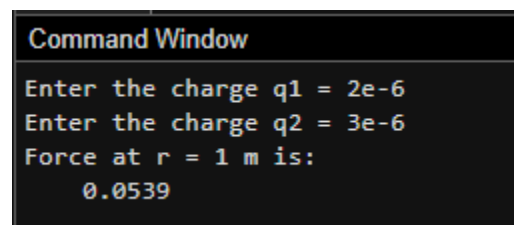
r = linspace(0.1,10,100);
F = k*q1*q2./(r.^2);

disp('Force at r = 1 m is:');
disp(k*q1*q2);

figure;
plot(r,F,'LineWidth',2);
grid on;
xlabel('Distance (m)');
ylabel('Force (N)');
title('Force between Two Electric Charges');
```

Command Prompt

```
Enter the charge q1 = 2e-6
Enter the charge q2 = 3e-6
Force at r = 1 m is:
    0.0539
```



Output Graph

